Theorem 1 (Asymptotic predictive risk bound for transfer with structured missing features)**.** *Fix a target site t and source sites $m = 1, \dots, M$. Let $S := \mathcal{R}(t) \subset [p]$, $q := |S|$, and write $S^c = [p] \setminus S$. Assume:*

(A1) ***Gaussian design.*** *For every site, $x \sim N(0, \Sigma)$, where $\Sigma \succeq 0$ is deterministic, $\lambda_{\min}(\Sigma) \geq c_\Sigma > 0$, $\lambda_{\max}(\Sigma) \leq C_\Sigma < \infty$, and the empirical spectral distribution of Σ_{SS} converges weakly to a compactly supported probability law H.*

(A2) ***Proportional asymptotics.*** *$q/n_t \to \gamma_t \in (0, \infty)$, and for each source site m, $q_m/n_m \to \gamma_m \in (0, \infty)$, where $q_m := |\mathcal{R}(m)|$. Also $n_m/n_t \to \tau_m \in (0, \infty)$.*

(A3) Common source parameter across source views. There is a single source coefficient vector $\beta^{(s)} \in \mathbb{R}^p$ such that for each source site m,

$$y^{(m)} = x^\top \beta^{(s)} + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2),$$

while the target site satisfies

$$y^{(t)} = x^\top \beta^{(t)} + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2).$$

(A4) Random effects on the target-observed regression problem. Define the target-observed optimal linear coefficient

$$\omega_*^{(t)} := \Phi^{(t)} \beta^{(t)} = \beta_S^{(t)} + \Sigma_{SS}^{-1} \Sigma_{SS^c} \beta_{S^c}^{(t)},$$

and the corresponding source anchor signal

$$w_* := \Phi^{(t)} \beta^{(s)} = \beta_S^{(s)} + \Sigma_{SS}^{-1} \Sigma_{SS^c} \beta_{S^c}^{(s)}.$$

Assume that, on $\mathbb{R}^q$,

$$\mathbb{E}[\omega_*^{(t)}] = 0, \qquad \mathrm{Var}(\omega_*^{(t)}) = \frac{\alpha_t^2}{q} I_q,$$

$$\mathbb{E}[w_*] = 0, \qquad \mathrm{Var}(w_*) = \frac{\alpha_s^2}{q} I_q,$$

and

$$\mathrm{Cov}(\omega_*^{(t)}, w_*) = \frac{\rho \alpha_t \alpha_s}{q} I_q, \qquad \rho \in [-1, 1].$$

(A5) ***Local ridge asymptotic linearity.*** *For each source site m, define the site-specific observed-feature oracle coefficient*

$$\omega_*^{(m)} := \Phi^{(m)} \beta^{(s)} = \beta_{\mathcal{R}(m)}^{(s)} + \Sigma_{\mathcal{R}(m),\mathcal{R}(m)}^{-1} \Sigma_{\mathcal{R}(m),\mathcal{R}(m)^c} \beta_{\mathcal{R}(m)^c}^{(s)}.$$

Let

$$\widehat{\omega}_{\mathrm{Loc}}^{(m)} := \arg\min_\omega \Big\{ \frac{1}{n_m} \|Y^{(m)} - X_{\mathcal{R}(m)}^{(m)} \omega\|_2^2 + \lambda_\omega^{(m)} \|\omega\|_2^2 \Big\}.$$

Assume that for each m,

$$\sqrt{n_t} \big(\widehat{\omega}_{\mathrm{Loc}}^{(m)} - \omega_*^{(m)} \big) \xrightarrow{d} N(0, V_m),$$

for a deterministic matrix $V_m \succeq 0$, where V_m depends on $\lambda_\omega^{(m)}$, γ_m, σ^2, $\Sigma_{\mathcal{R}(m),\mathcal{R}(m)}$, and the local random-effects signal strength through the ridge deterministic equivalent.

(A6) ***Aggregation identifiability.*** *Let*

$$G(W) := \sum_{m=1}^{M} (\Phi^{(m)})^\top W_m \Phi^{(m)}.$$

Assume that for the admissible weights considered below, $G(W)$ is invertible.

(A7) ***Transfer estimator.*** *Define the aggregated source estimator*

$$\widehat{\beta}^{(s)}(W) := \arg\min_{\beta \in \mathbb{R}^p} \sum_{m=1}^{M} \big\| \Phi^{(m)} \beta - \widehat{\omega}_{\mathrm{Loc}}^{(m)} \big\|_{W_m}^2.$$

Define the target anchor

$$\widehat{w}(W) := \Phi^{(t)} \widehat{\beta}^{(s)}(W) \in \mathbb{R}^q.$$

The target L_2-transfer estimator is

$$\widehat{\omega}_{\mathrm{TL}}^{(t)} := \arg\min_{\omega \in \mathbb{R}^q} \Big\{ \frac{1}{n_t} \|Y^{(t)} - X_S^{(t)} \omega\|_2^2 + \lambda_\delta \|\omega - \widehat{w}(W)\|_2^2 \Big\}.$$

(A8) ***Anchor-error spectral regularity.*** *There exists a deterministic matrix $C_A \succeq 0$ such that*

$$n_t \operatorname{Var}(\widehat{w}(W) - w_*) \to C_A,$$

and if $C_L := \lambda_{\min}(C_A)$, $C_U := \lambda_{\max}(C_A)$, then $0 \le C_L \le C_U < \infty$.

Then the following statements hold.

(i) Optimal aggregation weight. *The aggregation solution has the explicit form*

$$\widehat{\beta}^{(s)}(W) = G(W)^{-1} \sum_{m=1}^{M} (\Phi^{(m)})^\top W_m \widehat{\omega}_{\mathrm{Loc}}^{(m)},$$

and

$$\sqrt{n_t} \big(\widehat{\beta}^{(s)}(W) - \beta^{(s)} \big) \xrightarrow{d} N\Big(0,\, \Gamma(W)\Big),$$

with

$$\Gamma(W) = G(W)^{-1} \Big(\sum_{m=1}^{M} (\Phi^{(m)})^\top W_m V_m W_m \Phi^{(m)} \Big) G(W)^{-1}.$$

Among all deterministic $W_m \succeq 0$, the covariance $\Gamma(W)$ is minimized in the Loewner order by the generalized least-squares choice

$$W_m^\star = V_m^\dagger,$$

where $V_m^\dagger$ denotes the Moore–Penrose inverse, and when each V_m is invertible, $W_m^\star = V_m^{-1}$. The minimal covariance is

$$\Gamma^\star = \Big(\sum_{m=1}^{M} (\Phi^{(m)})^\top V_m^{-1} \Phi^{(m)} \Big)^{-1}.$$

(ii) Anchor error covariance. *For any admissible W,*

$$\sqrt{n_t} \big(\widehat{w}(W) - w_* \big) \xrightarrow{d} N\Big(0,\, C_A(W)\Big), \qquad C_A(W) := \Phi^{(t)} \Gamma(W) \Phi^{(t)\top},$$

and under the optimal aggregation weight,

$$C_A^\star = \Phi^{(t)}\Big(\sum_{m=1}^{M}(\Phi^{(m)})^\top V_m^{-1}\Phi^{(m)}\Big)^{-1}\Phi^{(t)\top}.$$

(iii) Deterministic-equivalent transfer-risk bound. *Let $v_t(z)$ be the companion Stieltjes transform associated with the limiting spectral law of $n_t^{-1}X_S^{(t)}X_S^{(t)\top}$, evaluated at $z \in \mathbb{C}\setminus\mathbb{R}_+$. Then, as $n_t, q \to \infty$,*

$$r^{\mathrm{pred}}\big(\widehat{\omega}_{\mathrm{TL}}^{(t)}\big) \implies R_{\mathrm{TL}}(\lambda_\delta),$$

and

$$R(\lambda_\delta, C_L) \;\le\; R_{\mathrm{TL}}(\lambda_\delta) \;\le\; R(\lambda_\delta, C_U),$$

where

$$R(\lambda, C) := \frac{\lambda^2(\alpha_t^2+\alpha_s^2+C-2\rho\alpha_t\alpha_s)\big[v_t(-\lambda)-\lambda v_t'(-\lambda)\big]}{\gamma_t\big[\lambda v_t(-\lambda)\big]^2} + \sigma^2\frac{v_t'(-\lambda)}{v_t(-\lambda)}.$$

(iv) Optimal transfer penalty. *For the upper bound $R(\lambda, C_U)$, the optimal penalty is*

$$\lambda_\delta^\star = \frac{\gamma_t\sigma^2}{\alpha_t^2+\alpha_s^2+C_U-2\rho\alpha_t\alpha_s},$$

and the corresponding optimal upper bound is

$$\inf_{\lambda_\delta>0} R(\lambda_\delta, C_U) = \sigma^2\lambda_\delta^\star v_t(-\lambda_\delta^\star).$$

If $C_A^\star = c_A I_q$ is asymptotically isotropic, then the bound is exact with $C_L = C_U = c_A$, and

$$R_{\mathrm{TL}}^\star = \sigma^2\lambda_\delta^\star v_t(-\lambda_\delta^\star), \qquad \lambda_\delta^\star = \frac{\gamma_t\sigma^2}{\alpha_t^2+\alpha_s^2+c_A-2\rho\alpha_t\alpha_s}.$$

(v) Choice of local ridge penalties. *For each source site m, the asymptotically optimal local penalty $\lambda_\omega^{(m)\star}$ is the minimizer of the site-wise ridge predictive risk for the projected regression problem*

$$Y^{(m)} = X_{\mathcal{R}(m)}^{(m)}\omega_*^{(m)} + \xi^{(m)},$$

with design covariance $\Sigma_{\mathcal{R}(m),\mathcal{R}(m)}$. Equivalently, $\lambda_\omega^{(m)\star}$ is the value minimizing the corresponding Dobriban–Wager deterministic equivalent for that local problem. With this choice, V_m is minimized, hence $C_A^\star$ is minimized, and therefore the transfer upper bound in (iii)–(iv) is minimized.

Proof. We divide the proof into five steps.

Step 1: Oracle coefficients under structured missingness. Fix a site $m \in \{t, 1, \dots, M\}$, and write $R_m := \mathcal{R}(m)$. Partition $x = (x_{R_m}^\top, x_{R_m^c}^\top)^\top$. Since $x \sim N(0, \Sigma)$, the conditional expectation of the unobserved coordinates given the observed ones is

$$\mathbb{E}[x_{R_m^c} \mid x_{R_m}] = \Sigma_{R_m^c,R_m}\Sigma_{R_m,R_m}^{-1}x_{R_m}.$$

Therefore

$$\mathbb{E}[y^{(m)} \mid x_{R_m}] = x_{R_m}^\top\beta_{R_m}^{(u)} + \mathbb{E}[x_{R_m^c}^\top\beta_{R_m^c}^{(u)} \mid x_{R_m}] = x_{R_m}^\top\Big(\beta_{R_m}^{(u)} + \Sigma_{R_m,R_m}^{-1}\Sigma_{R_m,R_m^c}\beta_{R_m^c}^{(u)}\Big),$$

where $u = s$ for source sites and $u = t$ for the target site. Hence the best linear predictor based only on x_{R_m} has coefficient

$$\omega_*^{(m)} = \Phi^{(m)}\beta^{(u)}.$$

In particular, for the target site,

$$\omega_*^{(t)} = \Phi^{(t)}\beta^{(t)}, \qquad w_* = \Phi^{(t)}\beta^{(s)}.$$

Step 2: Exact aggregation formula and optimal weights. By definition,

$$\widehat{\beta}^{(s)}(W) = \arg\min_{\beta} \sum_{m=1}^{M} \big(\Phi^{(m)}\beta - \widehat{\omega}_{\mathrm{Loc}}^{(m)}\big)^\top W_m \big(\Phi^{(m)}\beta - \widehat{\omega}_{\mathrm{Loc}}^{(m)}\big).$$

The objective is quadratic in β, so the first-order condition is

$$\sum_{m=1}^{M} (\Phi^{(m)})^\top W_m \Phi^{(m)} \widehat{\beta}^{(s)}(W) = \sum_{m=1}^{M} (\Phi^{(m)})^\top W_m \widehat{\omega}_{\mathrm{Loc}}^{(m)}.$$

Under (A6), $G(W)$ is invertible, hence

$$\widehat{\beta}^{(s)}(W) = G(W)^{-1} \sum_{m=1}^{M} (\Phi^{(m)})^\top W_m \widehat{\omega}_{\mathrm{Loc}}^{(m)}.$$

Because $\omega_*^{(m)} = \Phi^{(m)}\beta^{(s)}$, we obtain the exact error decomposition

$$\widehat{\beta}^{(s)}(W) - \beta^{(s)} = G(W)^{-1} \sum_{m=1}^{M} (\Phi^{(m)})^\top W_m \big(\widehat{\omega}_{\mathrm{Loc}}^{(m)} - \omega_*^{(m)}\big).$$

Multiplying by $\sqrt{n_t}$ and using (A5),

$$\sqrt{n_t}\big(\widehat{\beta}^{(s)}(W) - \beta^{(s)}\big) \xrightarrow{d} N(0, \Gamma(W)),$$

with

$$\Gamma(W) = G(W)^{-1} \Big(\sum_{m=1}^{M} (\Phi^{(m)})^\top W_m V_m W_m \Phi^{(m)} \Big) G(W)^{-1}.$$

We now optimize W. This is a standard generalized least-squares problem. Let $e_m := \widehat{\omega}_{\mathrm{Loc}}^{(m)} - \omega_*^{(m)}$, and stack

$$e = (e_1^\top, \ldots, e_M^\top)^\top, \qquad A = \begin{bmatrix} \Phi^{(1)} \\ \vdots \\ \Phi^{(M)} \end{bmatrix}, \qquad \Omega = \mathrm{diag}(V_1, \ldots, V_M).$$

Then the asymptotic linear model is

$$\widehat{\omega}_{\mathrm{stack}} = A\beta^{(s)} + n_t^{-1/2} e + o_p(n_t^{-1/2}),$$

with covariance Ω. The Aitken theorem implies that the linear unbiased estimator with minimum asymptotic covariance is the GLS estimator with block weight $\Omega^\dagger$, that is,

$$W_m^\star = V_m^\dagger.$$

If each V_m is invertible, $W_m^\star = V_m^{-1}$. Substituting gives

$$\Gamma^\star = \Big(\sum_{m=1}^{M} (\Phi^{(m)})^\top V_m^{-1} \Phi^{(m)}\Big)^{-1}.$$

This proves (i).

Step 3: Distribution of the target anchor. Define

$$\widehat{w}(W) := \Phi^{(t)} \widehat{\beta}^{(s)}(W).$$

Because $w_* = \Phi^{(t)}\beta^{(s)}$,

$$\widehat{w}(W) - w_* = \Phi^{(t)}\big(\widehat{\beta}^{(s)}(W) - \beta^{(s)}\big).$$

Hence, by continuous mapping,

$$\sqrt{n_t}\big(\widehat{w}(W) - w_*\big) \xrightarrow{d} N(0, C_A(W)),$$

with

$$C_A(W) = \Phi^{(t)} \Gamma(W) \Phi^{(t)\top}.$$

Under $W = W^\star$,

$$C_A^\star = \Phi^{(t)} \Big(\sum_{m=1}^{M} (\Phi^{(m)})^\top V_m^{-1} \Phi^{(m)}\Big)^{-1} \Phi^{(t)\top}.$$

This proves (ii).

Step 4: Exact algebraic form of the transfer estimator. Let $X := X_S^{(t)} \in \mathbb{R}^{n_t \times q}$, $Y := Y^{(t)}$, and define

$$\widehat{\omega}_{\mathrm{TL}}^{(t)} = \arg\min_{\omega} \Big\{ \frac{1}{n_t} \|Y - X\omega\|_2^2 + \lambda_\delta \|\omega - \widehat{w}\|_2^2 \Big\}.$$

The first-order condition yields

$$\Big(\frac{1}{n_t} X^\top X + \lambda_\delta I_q\Big) \widehat{\omega}_{\mathrm{TL}}^{(t)} = \frac{1}{n_t} X^\top Y + \lambda_\delta \widehat{w}.$$

Let

$$S_t := \frac{1}{n_t} X^\top X.$$

Since

$$Y = X\omega_*^{(t)} + \varepsilon_t,$$

we obtain

$$\widehat{\omega}_{\mathrm{TL}}^{(t)} - \omega_*^{(t)} = (S_t + \lambda_\delta I_q)^{-1} \Big(\frac{1}{n_t} X^\top \varepsilon_t - \lambda_\delta (\omega_*^{(t)} - \widehat{w}) \Big).$$

Therefore, conditioning on X,

$$\begin{aligned} r^{\mathrm{pred}}\big(\widehat{\omega}_{\mathrm{TL}}^{(t)}\big) &= \mathbb{E}\Big[(\tilde{y}^{(t)} - \tilde{x}_S^\top \widehat{\omega}_{\mathrm{TL}}^{(t)})^2 \;\Big|\; X \Big] \\ &= \sigma^2 + \mathbb{E}\Big[(\widehat{\omega}_{\mathrm{TL}}^{(t)} - \omega_*^{(t)})^\top \Sigma_{SS} (\widehat{\omega}_{\mathrm{TL}}^{(t)} - \omega_*^{(t)}) \;\Big|\; X \Big]. \end{aligned}$$

Thus the predictive risk is exactly the noise floor σ^2 plus the target-observed parameter estimation error measured in the Σ_{SS}-metric.

Step 5: Deterministic equivalent and optimization of λ_δ. By Step 3, the target anchor can be written as

$$\widehat{w} = w_* + \delta_n, \qquad \delta_n := \widehat{w} - w_*,$$

with asymptotic covariance $n_t \operatorname{Var}(\delta_n) \to C_A$ by (A8). Under (A1), (A2), (A4), and (A8), the transfer problem on the target-observed coordinates is exactly the high-dimensional L_2-transfer ridge problem with source anchor $w_* + \delta_n$. Hence the deterministic-equivalent analysis for L_2-distance-based transfer applies, yielding the asymptotic bound

$$R(\lambda_\delta, C_L) \le R_{\mathrm{TL}}(\lambda_\delta) \le R(\lambda_\delta, C_U),$$

where

$$R(\lambda, C) = \frac{\lambda^2(\alpha_t^2 + \alpha_s^2 + C - 2\rho\alpha_t\alpha_s)\big[v_t(-\lambda) - \lambda v_t'(-\lambda)\big]}{\gamma_t[\lambda v_t(-\lambda)]^2} + \sigma^2 \frac{v_t'(-\lambda)}{v_t(-\lambda)}.$$

This proves (iii).

For the optimization, set

$$D(C) := \alpha_t^2 + \alpha_s^2 + C - 2\rho\alpha_t\alpha_s.$$

The corresponding optimal transfer penalty is

$$\lambda_\delta^\star(C) = \frac{\gamma_t\sigma^2}{D(C)},$$

and the minimized risk bound equals

$$\inf_{\lambda>0} R(\lambda, C) = \sigma^2 \lambda_\delta^\star(C)\, v_t(-\lambda_\delta^\star(C)).$$

Applying this with $C = C_U$ proves (iv). If $C_A^\star = c_A I_q$, then $C_L = C_U = c_A$ and the lower and upper bounds coincide, so the formula is exact.

Finally, $C_A^\star$ depends on the local penalties only through the local asymptotic covariances V_m. Therefore minimizing each V_m over $\lambda_\omega^{(m)}$ minimizes the GLS aggregation covariance $\Gamma^\star$, hence minimizes $C_A^\star$, hence minimizes the transfer bound. By the ridge deterministic-equivalent theory for each projected local regression problem, the optimal $\lambda_\omega^{(m)}$ is precisely the minimizer of the site-wise ridge predictive risk. This proves (v), and completes the proof. □